

An incoherent feed-forward loop mediates robustness and tunability in a plant immune network

Akira Mine^{1,2}, Tatsuya Nobori^{1,†}, Maria C Salazar-Rondon^{1,†}, Thomas M Winkelmüller¹, Shajahan Anver¹, Dieter Becker¹ & Kenichi Tsuda^{1,*} 

Abstract

Immune signaling networks must be tunable to alleviate fitness costs associated with immunity and, at the same time, robust against pathogen interferences. How these properties mechanistically emerge in plant immune signaling networks is poorly understood. Here, we discovered a molecular mechanism by which the model plant species *Arabidopsis thaliana* achieves robust and tunable immunity triggered by the microbe-associated molecular pattern, flg22. Salicylic acid (SA) is a major plant immune signal molecule. Another signal molecule jasmonate (JA) induced expression of a gene essential for SA accumulation, *EDS5*. Paradoxically, JA inhibited expression of *PAD4*, a positive regulator of *EDS5* expression. This incoherent type-4 feed-forward loop (I4-FFL) enabled JA to mitigate SA accumulation in the intact network but to support it under perturbation of *PAD4*, thereby minimizing the negative impact of SA on fitness as well as conferring robust SA-mediated immunity. We also present evidence for evolutionary conservation of these gene regulations in the family *Brassicaceae*. Our results highlight an I4-FFL that simultaneously provides the immune network with robustness and tunability in *A. thaliana* and possibly in its relatives.

Keywords incoherent feed-forward loop; jasmonate; plant immunity; salicylic acid; signaling perturbation

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Introduction

Proper processing of signals through signaling networks is central for organisms to respond accordingly to the signals. As such, signaling networks are comprised of recurring regulatory subnetwork structures called network motifs with various information-processing functions. Feed-forward loop (FFL), which consists of two regulators and

a target, represents a major class of network motifs [1]. Each of interactions among the components of a FFL can be either positive (activation) or negative (repression). As a result, there are eight possible structural configurations of FFL. Of these configurations, incoherent type-4 FFL (I4-FFL), in which a regulator has a positive effect on the target but a negative effect on the other regulator that positively regulates the target, is rare in biological networks and, therefore, its biological function has rarely been described.

In nature, plants are in constant contact with a wide variety of microbes, which often produce common molecular signatures known as microbe-associated molecular patterns (MAMPs) [2]. Plants sense MAMPs by plasma membrane-localized pattern recognition receptors and feed this information into signaling networks that finely control the output immune reaction designated as pattern-triggered immunity (PTI) [2–5]. Since recognized MAMPs are often common to a class of microbes [2], PTI could be triggered by both pathogenic and non-pathogenic microbes. Therefore, it is vital for plants to avoid unnecessary PTI against non-pathogenic microbes, as there is a trade-off between immunity and growth [6–9]. At the same time, it is important to retain PTI that is effective against pathogens that deploy virulence effectors to interfere with immune signaling components [10,11] and that can function under perturbation due to diverse environmental conditions [12]. The molecular mechanisms that allow these properties to emerge from PTI signaling networks are poorly understood.

Plants rely on PTI to resist necrotrophs that actively kill hosts to acquire nutrients as well as to resist biotrophs that require living hosts for multiplication [2,13]. The phytohormone jasmonate (JA) is a major contributor to immunity against necrotrophs [13]. JA is produced in response to MAMPs such as flg22 [14] and chitin [15], a part of bacterial flagellin and a part of fungal cell walls, respectively. JA biosynthesis requires allene oxide synthase encoded by *DELAYED-DEHISCENCE 2 (DDE2)* [16]. JA and its derivatives including methyl JA (MeJA) can be converted to JA-isoleucine (JA-Ile) [17,18]. Perception of JA-Ile by the F-box protein CORONATINE INSENSITIVE 1 (COI1) leads to ubiquitination- and proteasome-dependent degradation of JASMONATE ZIM DOMAIN

¹ Department of Plant Microbe Interactions, Max Planck Institute for Plant Breeding Research, Cologne, Germany

² Center for Gene Research, Nagoya University, Chikusa-Ku, Nagoya, Japan

*Corresponding author. Tel: +49 221 5062 302; E-mail: tsuda@mpipz.mpg.de

[†]These authors contributed equally to this work